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***EE321: Computer Architecture***
***Control Solution (1h)***
***2022 – 2023***

**Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.**

## 1. Simple CPU Function and Structure (15 pts)

Consider a Little-Endian simple 8-bit CPU with the following specifications: 16-bit address bus, the memory is byte addressable, the instruction list and opcodes are given in Table 1, the instruction formats are given in Table 2. The CPU has 8 data registers (R0 to R7) and 8 address registers (A0 to A7), the register codes are given in Table 3. A0 is the program counter and A7 is the stack pointer. The number of clock cycles for each basic cycle are given in Table 4.

1.1 Give the instructions that are using: a) direct addressing, and b) indirect addressing.

*a) Direct addressing: LDD, LDR, STD and ADD.*
*b) Indirect addressing: None.* *(1,5 pt.)*

1.2 Give the size in bits of the registers: R0-R7, A0-A7, MAR, and IR. Justify.

*a) R0-R7: 8 bits, they are data registers and the CPU is an 8-bit processor.*
*b) A0-A7, MAR: 16 bits, they are address registers and the CPU has 16-bit address bus*
*c) IR: 24 bits, the size of the biggest instruction is 3 bytes.* *(1,5 pt.)*

1.3 Give the size in bytes of the instructions: a) LDD Rd,(Y), b) INC Ad, and c) LDR Ad,(As). Justify.

a) LDD Rd,(Y): 3 bytes, 1 byte for the opcode and 2 bytes for the operand Y
b) INC Ad: 1 byte, the operand is coded within the opcode.
c) LDR Ad,(As): 2 bytes, 1 byte for the opcode and 1 byte for operand As. *(1,5 pt.)*

1.4 What is the difference between the two instructions: LDR Rd,(As) and LDR Ad,(As). How the CPU makes the difference between the two instructions.

LDR Rd,(As) loads 1-byte from the memory and LDR Ad,(As) loads two bytes from the memory. The two instructions have different register code. *(1,5 pt.)*

1.5 Using the same format as in Table 1, give the description of the instructions: a) POP Rd, and b) POP Ad.

*a) POP Rd: Rd = [A7] and A7=A7+1*
*b) POP Ad: $Ad_{7...0}$ = [A7], $Ad_{15...8}$ = [A7+1], A7 = A7+2* *(1,5 pt.)*

1.6 Assuming all the registers are connected together using internal bata bus, give the micro-operations of the **execute** cycle of the instructions: a) MOV Ad,Y, and b) LDR Rd,(As).

a) t1: Ad <--- IR(Y)
b) t1: MAR <--- As
t2 : MBR <--- Memory
t3: Rd <--- MBR *(1,5 pt.)*

1.7 Give the HEX code of the instructions: a) ADI R1,10H, b) ADI A1,78H, and c) INC A1.

*a) ADI R1, 10H: 1000 0001 0001 0002 = 8110H*
*b) ADI A1,78H: 1000 1001 0000 0000 0011 0100 = 897800H*
*c) INC A1: 1100 1001 = C9H* *(1,5 pt.)*

1.8 Give the HEX content of the registers A0 (PC), A1, MAR, and IR after the execution of the instruction LDD A1, (F122H) stored at address 89A8H.

A0 = 89ABH
$A1_{7..0}$ = [F122H] , $A1_{15..8}$ = [F123H]
MAR = F123H
IR = 49F122H *(2 pts.)*

1.9 Which of the following states: IAC, IF, IOD, OAC, OF, DO, and OS, are involved in the instructions: a) PUSH Ad, and b) POP Ad?

a) PUSH Ad: IAC, IF, IOD, OAC, and OS.
b) POP Ad: IAC, IF, IOD, OAC, and OF. *(1,5 pt.)*

1.10 Give the number of clock cycles it takes to execute (i.e. fetch and execute) the instructions: a) LDR Rd, (As), b) LDD Ad,(Y), and c) PUSH Rd . Justify.

a) LDR Rd,(As): 10 clock cycles, 4 for opcode fetch, 3 for operand fetch, and 3 for data read.
b) LDD Ad,(Y): 16 clock cycles, 4 for opcode fetch, 6 for operand fetch, and 6 for data read.
c) PUSH Rd: 7 clock cycles, 4 for opcode fetch, and 3 for data store. *(1,5 pt.)*

## 2. Basic Concepts ( 5 pts)

2.1 Consider the 2's complement representation using 16 bits, complete the following equalities:

*a) $(7F00H)_{2'C} = (32512)_{10}$* *$(FF00H)_{2'C} = (-256)_{10}$*
*b) $(+257)_{10} = (0101H)_{2'C}$* *$(-257)_{10} = (FEFFH)_{2'C}$* *(1,5 pt.)*

2.2 Consider a microprogrammed control unit with horizontal microinstructions. Assuming 128 control signals (internal and external), 500 microinstructions, and 3 bits for jump condition field, give the number of bits of the jump address field. Justify.

*We have 500 microinstructions (locations), so the number of bits of the jump address field is 9.* *(1,5 pt.)*

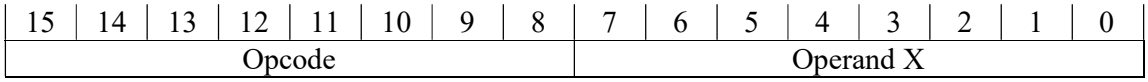
2.3 Consider a CPU with a 20-bit address bus and 16-bit data bus. Assuming the memory range ABC00H-ABDF0H of the CPU is mapped to a RAM memory chip, what is the size in bits and the number of address lines of the RAM.

*a) Number of address lines: 9*
*b) Size = $2^9$ * 16 = 8196 bits* *(2 pts.)*

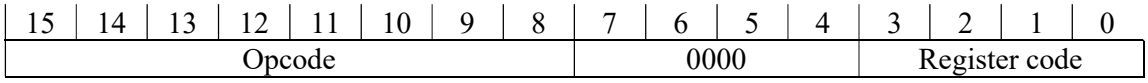
| Instruction | Opcode (binary) | Instruction Size | Description |
|---|---|---|---|
| MOV Rd, X | $0001C_3C_2C_1C_0$ | 2 bytes | Rd = X |
| MOV Ad, Y | $0001C_3C_2C_1C_0$ | 3 bytes | Ad = Y |
| MOV Rd, Rs | $0010C_3C_2C_1C_0$ | 2 bytes | Rd = Rs |
| PUSH Rd | $0011C_3C_2C_1C_0$ | 1 byte | A7 = A7-1, [A7] = Rd |
| PUSH Ad | $0011C_3C_2C_1C_0$ | | |
| LDD Rd, (Y) | $0100C_3C_2C_1C_0$ | | Rd = [Y] |
| LDR Rd, (As) | $0101C_3C_2C_1C_0$ | | Rd = [As] |
| LDR Ad, (As) | $0101C_3C_2C_1C_0$ | | Ad = [As] |
| STD (Y), Rd | $0110C_3C_2C_1C_0$ | | [Y] = Rd |
| STD (Ad), Rs | $0111C_3C_2C_1C_0$ | | [Ad] = Rs |
| ADI Rd, X | $1000C_3C_2C_1C_0$ | | Rd = Rd + X |
| ADI Ad, Y | $1000C_3C_2C_1C_0$ | | Ad = Ad + Y |
| ADD Rd, (As) | $1001C_3C_2C_1C_0$ | | Rd = Rd + [As] |
| ADR Rd, Rs<br>ADR Ad, As | $1010C_3C_2C_1C_0$ | | Rd = Rd + Rs<br>Ad = Ad + As |
| POP Rd<br>POP Ad | $1011C_3C_2C_1C_0$ | | |
| INC Rd<br>INC Ad | $1100C_3C_2C_1C_0$ | | Rd = Rd + 1<br>Ad = Ad + 1 |
| HLT | 1111000 | | Stop the execution of the program |

*Rd =Data register (it can be any data register).*
*Rs = Data register (it can be any data register).*
*Ad = Address register (it can be any address register).*
*As = Address register (it can be any address register).*
*$C_3C_2C_1C_0$ = Binary code of the operand (Rd / Ad) (see Table 3).*
*X =8-bit 2's complement value.*
*Y =16-bit 2's complement value.*
*[…] = Content of memory location.*

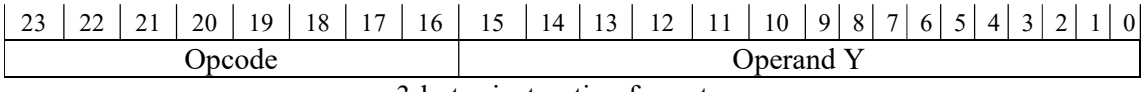
**Table 1: Instructions description.**



2-bytes instruction format.



2-bytes instruction format.



3-bytes instruction format

**Table 2: 2-bytes and 3-bytes instructions formats.**

| Register | R0 | R1 | R2 | R3 | R4 | R5 | R6 | R7 |
|---|---|---|---|---|---|---|---|---|
| Code | 0000 | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 |

| Register | A0 | A1 | A2 | A3 | A4 | A5 | A6 | A7 |
|---|---|---|---|---|---|---|---|---|
| Code | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |

**Table 3: Binary codes of data and address registers.**

| Cycle | Number of clock cycles |
|---|---|
| Opcode fetch | 4 clock cycles |
| 1-byte operand fetch | 3 clock cycles |
| Memory read | 3 clock cycles |
| Memory write | 3 clock cycles |
| Do operation | 1 clock cycle |

**Table 4: Number of clock cycle for each basic cycle.**